OOAD

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Class Test

1. Architectural Patterns & MVC

An architectural pattern is a reusable structure for organizing a software system. Two patterns are MVC and Three-Tier Architecture

MVC has:

* Model - manages data
* View - displays data.
* Controller - handles user requests.

It improves modularity because each part has a separate responsibility, making the system easier to maintain

2. DAO Pattern

DAO (Data Access object) separates database operations from application logic.

* Easy to maintain.
* Reduces database dependency in business logic

~~It~~ handles operations like insert, update, delete, and retrieve, while the application focuses on its main logic

3. Design Guidelines & LSP

object oriented design guidelines are principles used to create reusable and maintainable software.

1. Class level guidelines
2. Relationship level guidelines.

4). Database Persistence & ORM

Persistence means storing object data permanently in a database

Two mechanisms are

1. ORM
2. Directed database access

ORM maps objects to database tables and helps store and retrieve objects easily without writing much database code.

5). Application Layer

The Application Layer coordinates the execution of use cases.

Two responsibilities:

1. Coordinate application operations
2. Manage the workflow of requests

6). Three Tier Architecture:

1. Presentation layer - user interface
2. Business Layer - business processing
3. Data layer - database operations.

In e-commerce:

Customer - Presentation - Business - Data - Database

7). Business Layer Logic

~~Tea~~ It contains business rules & processes requests.

two functions:

1. Validate requests
2. Process business operations

Eg: In banking, it checks whether sufficient balance exists before processing a withdrawal.

8. Cohesion & Coupling

Cohesion is how closely related the responsibilities of a class are.

Coupling is the dependency between classes.

* High cohesion - focused classes
* Low coupling - fewer dependencies

9. Encapsulation:

Encapsulation means combining data and methods inside a class and controlling access to the data.

Eg: Class Account {
        private double balance;
    }

10). Online booking Layered System:

Major Layers: (i) Presentation Layer
(ii) Application Layer.

3. Business Layer

4. Data Access Layer

5. Infrastructure Layer

Useful patterns are MVC, DAO, and Dependency Layer.

11). Infrastructure Layer:

The Infrastructure Layer provides technical services to the application.

Two responsibilities:

1. Database access
2. External system communication

12) Singleton Pattern:

Singleton pattern ensures that only one object of a class is created.

It has a private constructor and provides controlled access to the object.

Use: Managing shared resources

13. Dependency Injection

Dependency Injection (DI) means providing required objects from outside the class.

1. Reduces tight coupling
2. Improves testing

14). Strategy pattern:

Strategy pattern allows different algorithm algorithms / behaviours to be selected at runtime

- Strategy : Defines the behaviour.
- Context : Uses the selected ~~strat~~ strategy

Eg: UPI, credit card.

15). Library Management System :

Layers:

1. Presentation - user interface
2. Application - coordinate operations
3. Business - handles library rules.
4. Data Access - manages database.
5. Infrastructure - technical services.

Patterns: MVC, DAO, Strategy, DI.

It improves flexibility and maintainability.